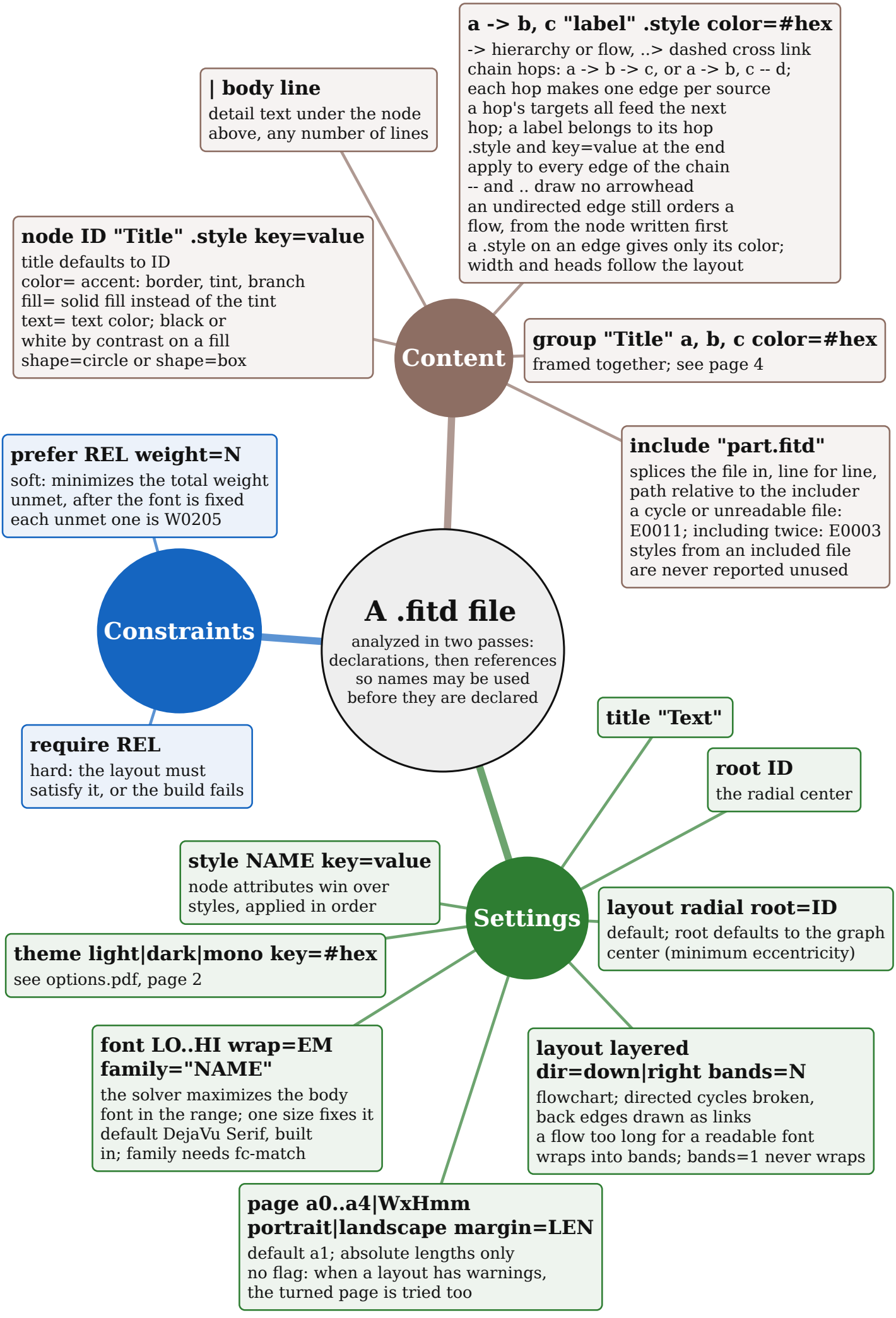
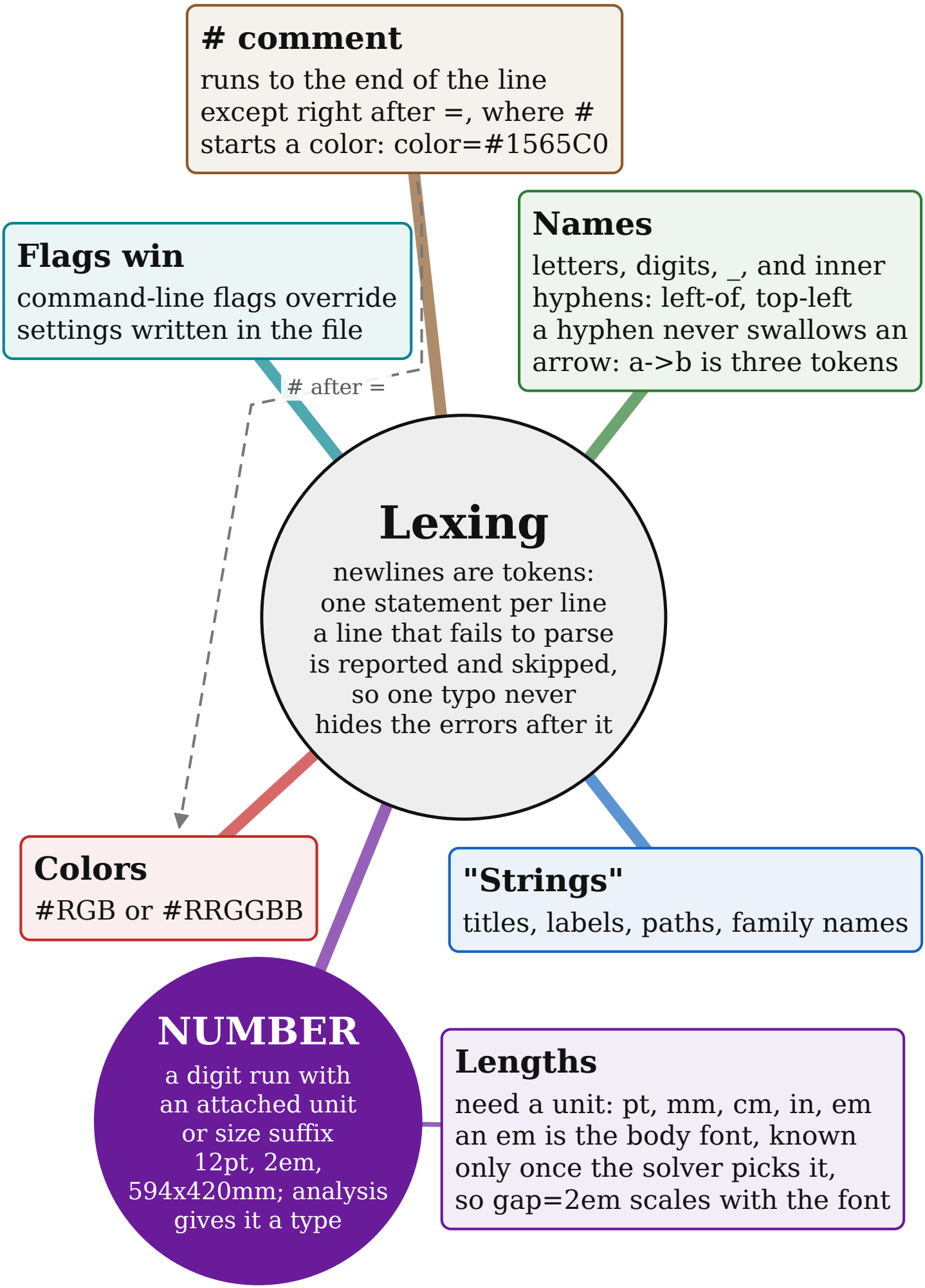
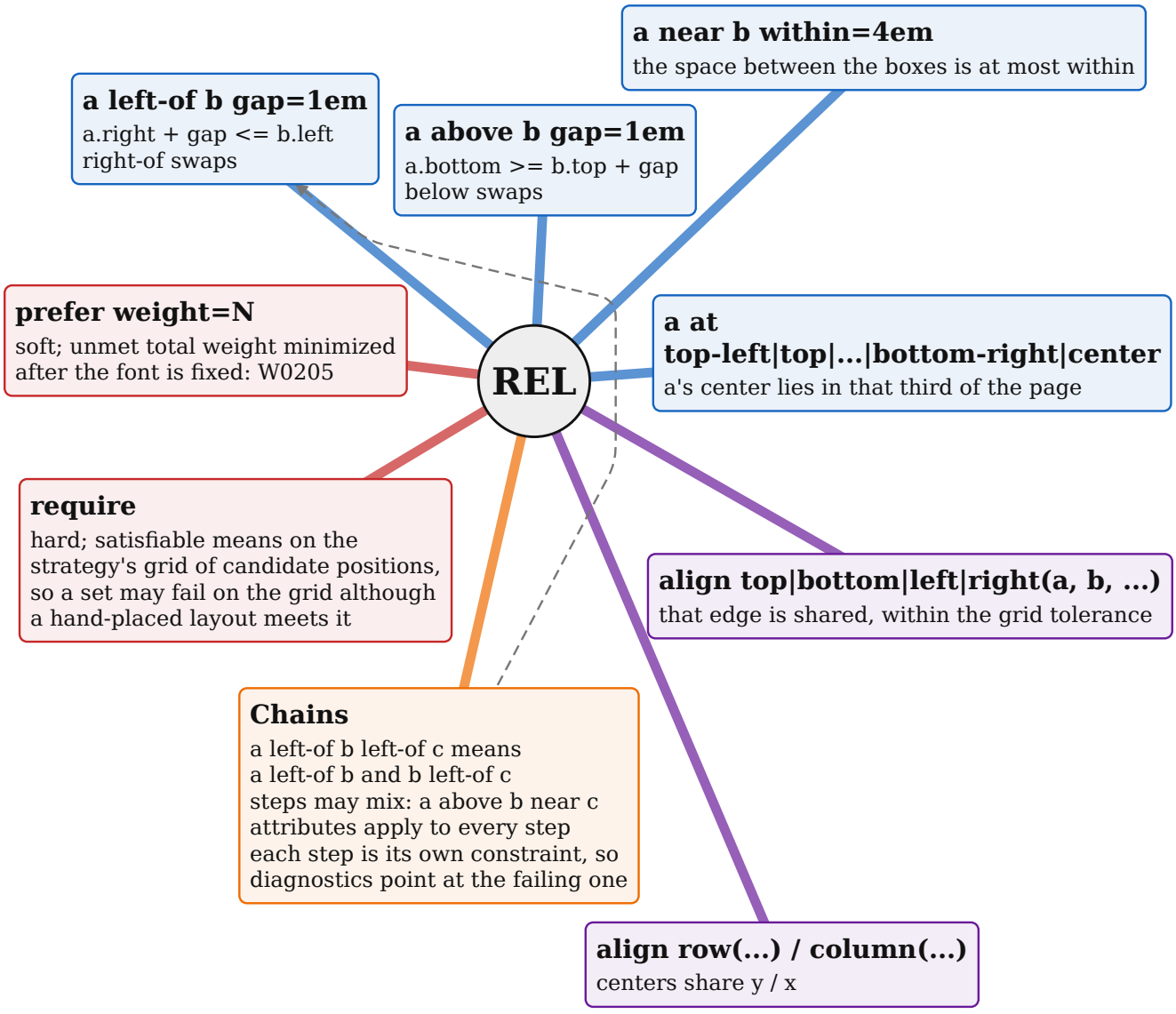
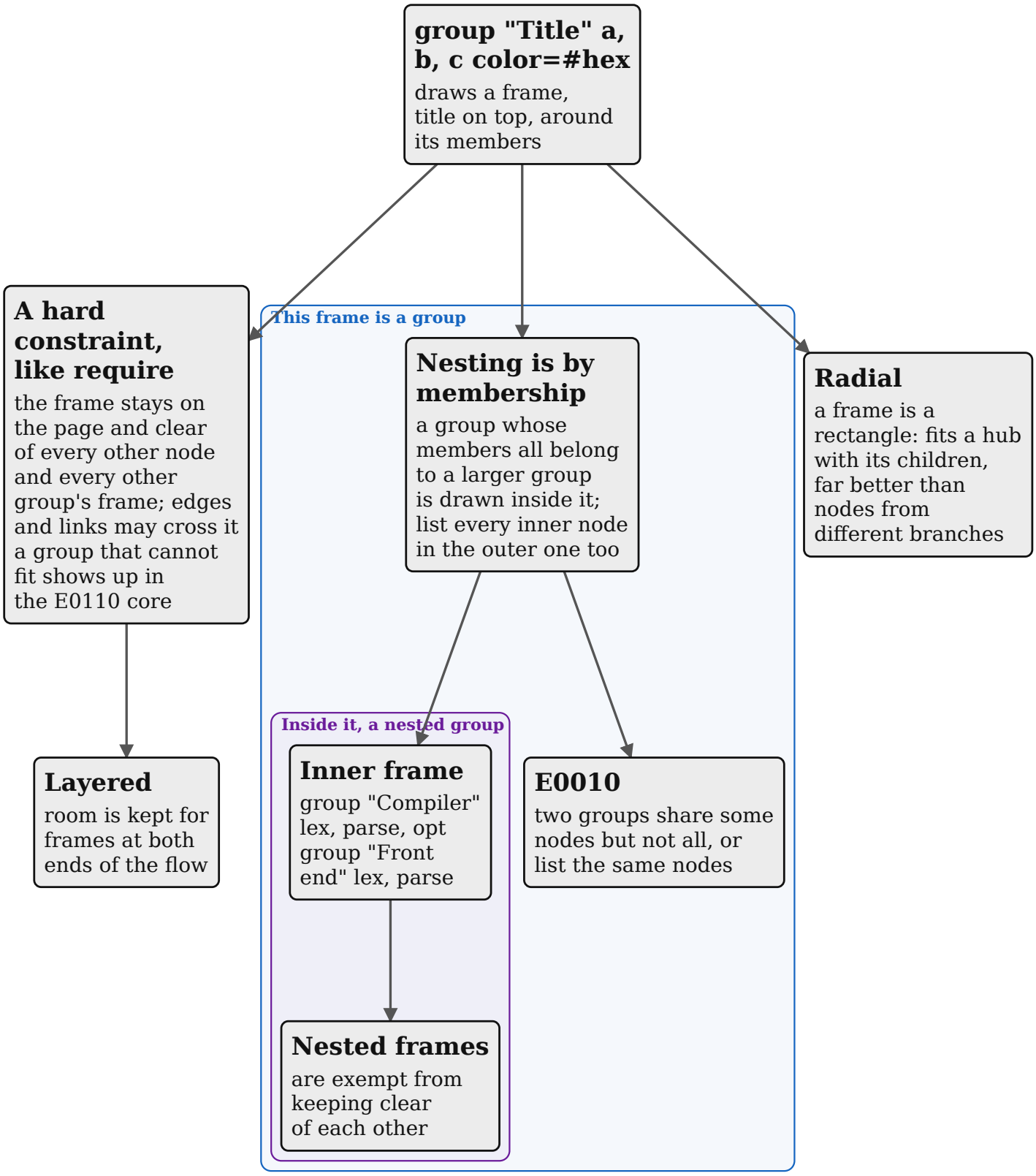
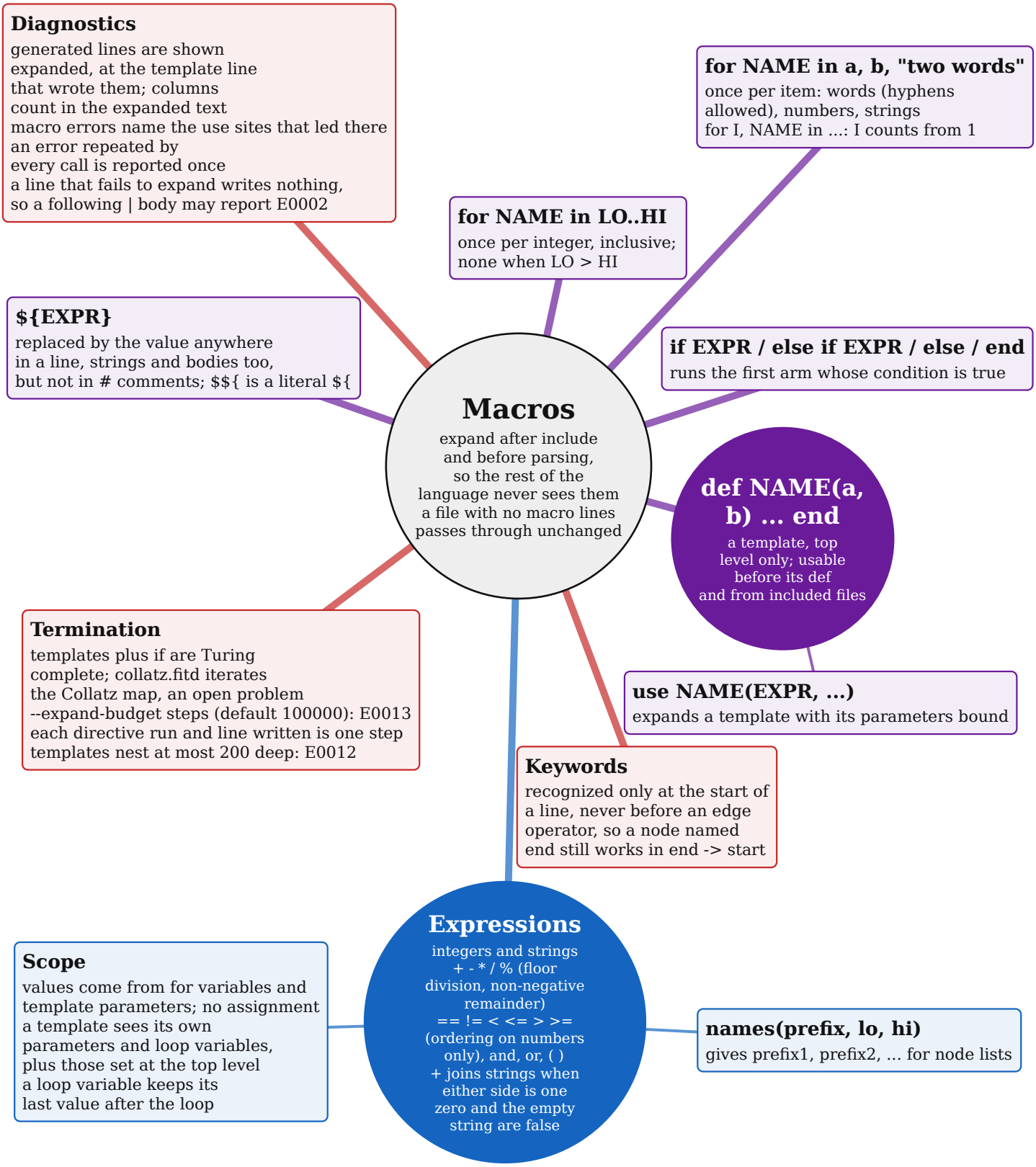
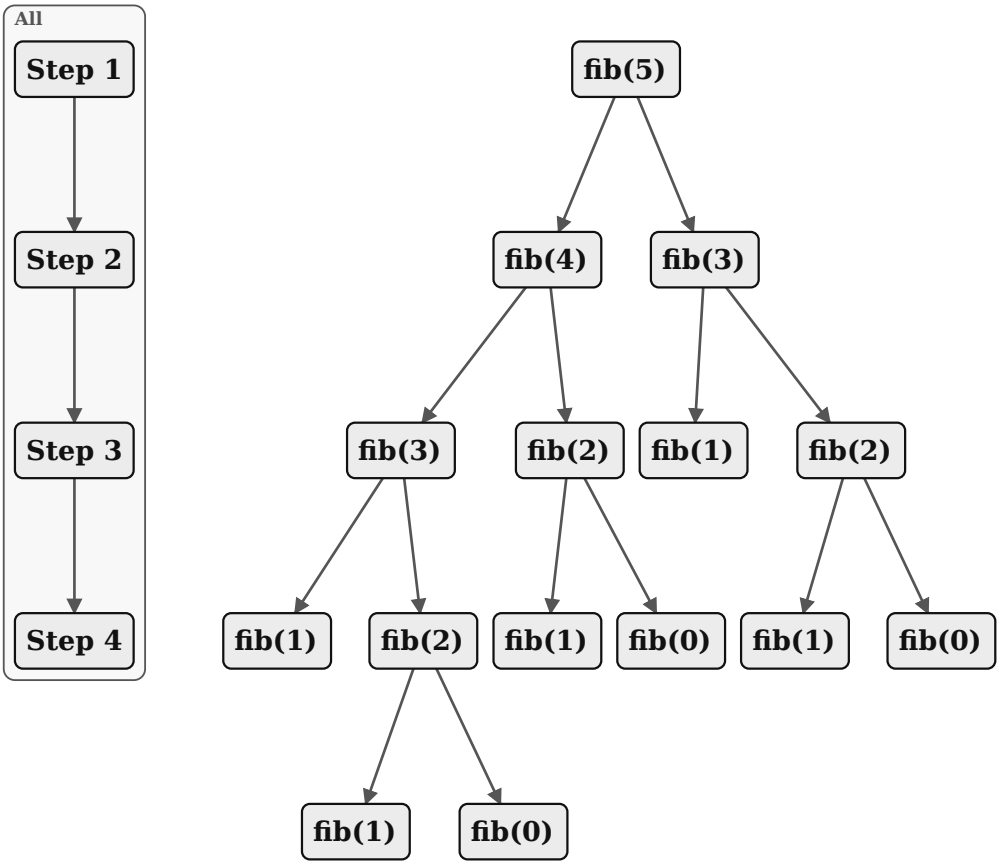

A .fitd file
analyzed in two passes: declarations, then references so names may be used before they are declared
Content
a -> b, c "label" .style color=#hex
-> hierarchy or flow, ..> dashed cross link
chain hops: a -> b -> c, or a -> b, c -- d;
each hop makes one edge per source
a hop's targets all feed the next hop; a label belongs to its hop
.style and key=value at the end apply to every edge of the chain
-- and .. draw no arrowhead
an undirected edge still orders a flow, from the node written first
a .style on an edge gives only its color; width and heads follow the layout
| body line
detail text under the node above, any number of lines
node ID "Title" .style key=value
title defaults to ID
color= accent: border, tint, branch
fill= solid fill instead of the tint
text= text color; black or white by contrast on a fill
shape=circle or shape=box
group "Title" a, b, c color=#hex
framed together; see page 4
include "part.fitd"
splices the file in, line for line, path relative to the includer
a cycle or unreadable file: E0011; including twice: E0003
styles from an included file are never reported unused
Constraints
prefer REL weight=N
soft: minimizes the total weight unmet, after the font is fixed
each unmet one is W0205
require REL
hard: the layout must satisfy it, or the build fails
Settings
title "Text"
root ID
the radial center
style NAME key=value
node attributes win over styles, applied in order
theme light|dark|mono key=#hex
see options.pdf, page 2
layout radial root=ID
default; root defaults to the graph center (minimum eccentricity)
font LO..HI wrap=EM family="NAME"
the solver maximizes the body font in the range; one size fixes it
default DejaVu Serif, built in; family needs fc-match
layout layered dir=down|right bands=N
flowchart; directed cycles broken, back edges drawn as links
a flow too long for a readable font wraps into bands; bands=1 never wraps
page a0..a4|WxHmm portrait|landscape margin=LEN
default a1; absolute lengths only
no flag: when a layout has warnings, the turned page is tried too

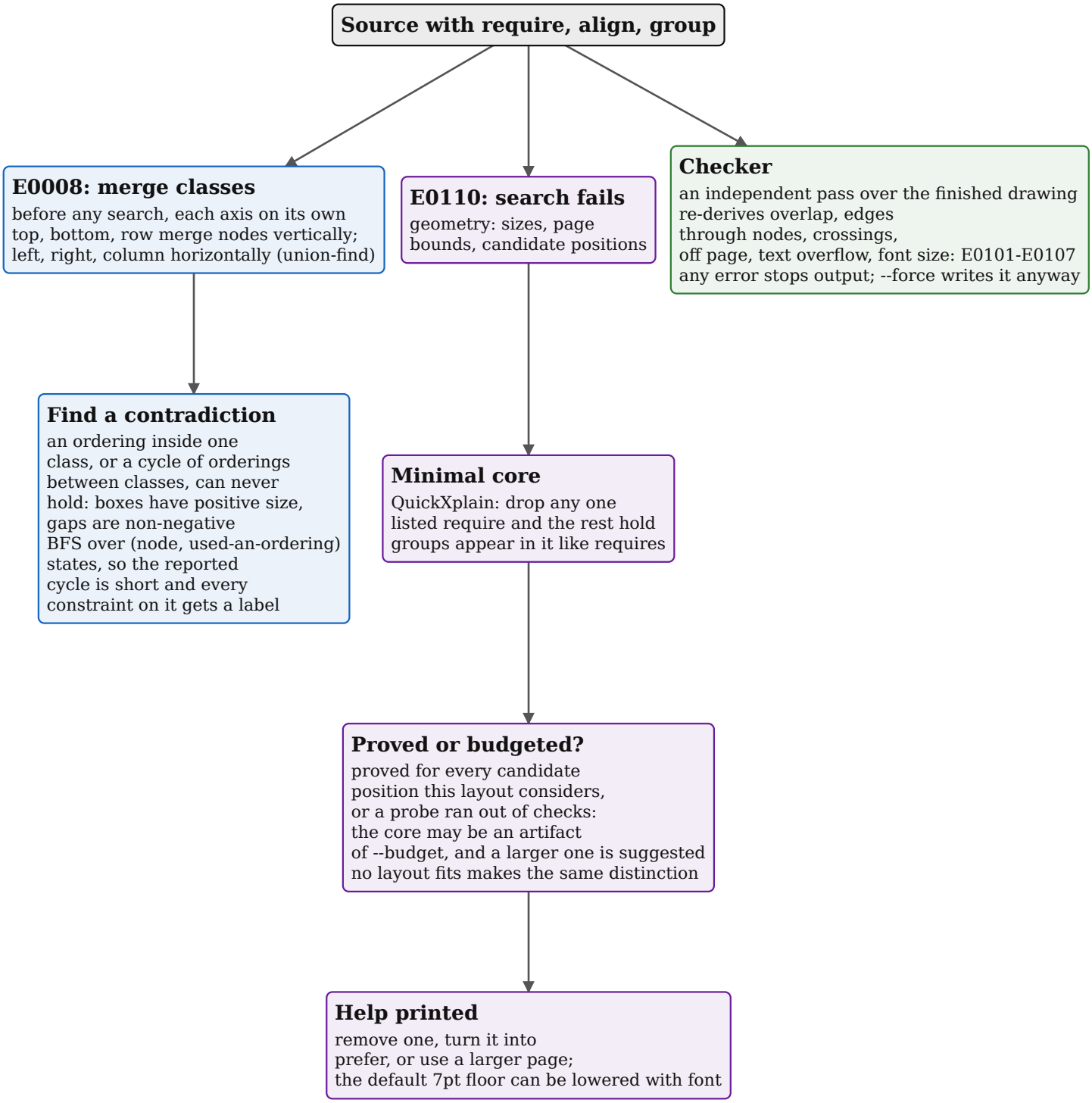

comment
runs to the end of the line
except right after =, where #
starts a color: color=#1565C0
Flags win
command-line flags override
settings written in the file
after =
Names
letters, digits, _, and inner
hyphens: left-of, top-left
a hyphen never swallows an
arrow: a->b is three tokens
Lexing
newlines are tokens:
one statement per line
a line that fails to parse
is reported and skipped,
so one typo never
hides the errors after it
Colors
#RGB or #RRGGBB
"Strings"
titles, labels, paths, family names
NUMBER
a digit run with
an attached unit
or size suffix
12pt, 2em,
594x420mm; analysis
gives it a type
Lengths
need a unit: pt, mm, cm, in, em
an em is the body font, known
only once the solver picks it,
so gap=2em scales with the font

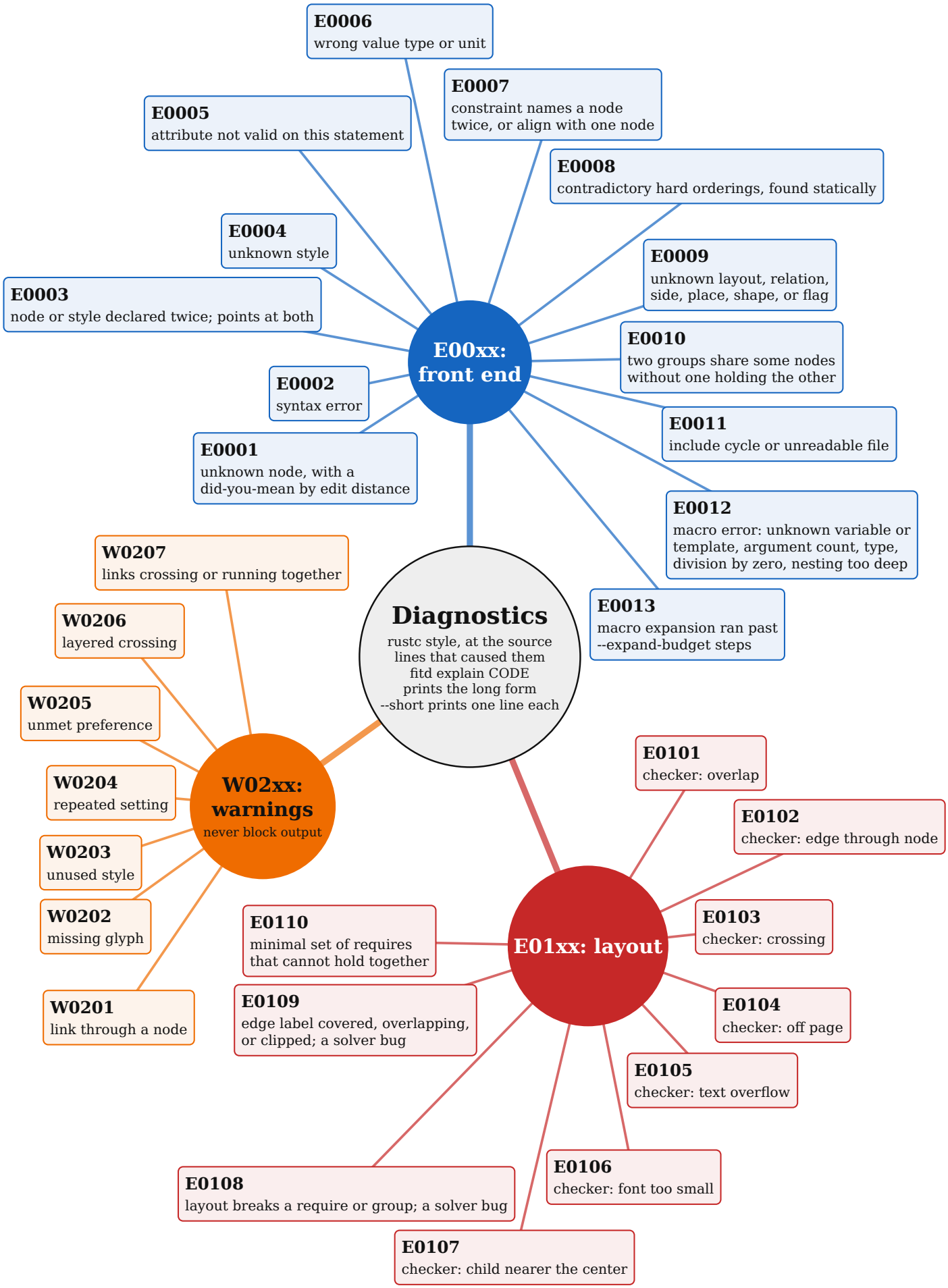

a near b within=4em
the space between the boxes is at most within
a left-of b gap=1em
a.right + gap <= b.left
right-of swaps
a above b gap=1em
a.bottom >= b.top + gap
below swaps
prefer weight=N
soft; unmet total weight minimized
after the font is fixed: W0205
REL
a at
top-left|top|...|bottom-right|center
a's center lies in that third of the page
require
hard; satisfiable means on the
strategy's grid of candidate positions,
so a set may fail on the grid although
a hand-placed layout meets it
align top|bottom|left|right(a, b, ...)
that edge is shared, within the grid tolerance
Chains
a left-of b left-of c means
a left-of b and b left-of c
steps may mix: a above b near c
attributes apply to every step
each step is its own constraint, so
diagnostics point at the failing one
align row(...) / column(...)
centers share y / x


group "Title" a, b, c color=#hex
draws a frame, title on top, around its members
A hard constraint, like require
the frame stays on the page and clear of every other node and every other group's frame; edges and links may cross it a group that cannot fit shows up in the E0110 core
This frame is a group
Nesting is by membership
a group whose members all belong to a larger group is drawn inside it; list every inner node in the outer one too
Radial
a frame is a rectangle: fits a hub with its children, far better than nodes from different branches
Inside it, a nested group
Inner frame
group "Compiler" lex, parse, opt group "Front end" lex, parse
E0010
two groups share some nodes but not all, or list the same nodes
Nested frames
are exempt from keeping clear of each other
Layered
room is kept for frames at both ends of the flow


Diagnostics
generated lines are shown
expanded, at the template line
that wrote them; columns
count in the expanded text
macro errors name the use sites that led there
an error repeated by
every call is reported once
a line that fails to expand writes nothing,
so a following | body may report E0002
for NAME in a, b, "two words"
once per item: words (hyphens
allowed), numbers, strings
for I, NAME in ...: I counts from 1
for NAME in LO..HI
once per integer, inclusive;
none when LO > HI
${EXPR}
replaced by the value anywhere
in a line, strings and bodies too,
but not in # comments; $${ is a literal ${
if EXPR / else if EXPR / else / end
runs the first arm whose condition is true
Macros
expand after include
and before parsing,
so the rest of the
language never sees them
a file with no macro lines
passes through unchanged
def NAME(a, b) ... end
a template, top
level only; usable
before its def
and from included files
Termination
templates plus if are Turing
complete; collatz.fitd iterates
the Collatz map, an open problem
--expand-budget steps (default 100000): E0013
each directive run and line written is one step
templates nest at most 200 deep: E0012
use NAME(EXPR, ...)
expands a template with its parameters bound
Keywords
recognized only at the start of
a line, never before an edge
operator, so a node named
end still works in end -> start
Expressions
integers and strings
+ - * / % (floor
division, non-negative
remainder)
== != < <= > >=
(ordering on numbers
only), and, or, ( )
+ joins strings when
either side is one
zero and the empty
string are false
Scope
values come from for variables and
template parameters; no assignment
a template sees its own
parameters and loop variables,
plus those set at the top level
a loop variable keeps its
last value after the loop
names(prefix, lo, hi)
gives prefix1, prefix2, ... for node lists


All
Step 1
Step 2
Step 3
Step 4
fib(5)
fib(4)
fib(3)
fib(3)
fib(2)
fib(1)
fib(2)
fib(1)
fib(2)
fib(1)
fib(0)
fib(1)
fib(0)
fib(1)
fib(0)


Source with require, align, group
E0008: merge classes
before any search, each axis on its own
top, bottom, row merge nodes vertically;
left, right, column horizontally (union-find)
E0110: search fails
geometry: sizes, page
bounds, candidate positions
Checker
an independent pass over the finished drawing
re-derives overlap, edges
through nodes, crossings,
off page, text overflow, font size: E0101-E0107
any error stops output; --force writes it anyway
Find a contradiction
an ordering inside one
class, or a cycle of orderings
between classes, can never
hold: boxes have positive size,
gaps are non-negative
BFS over (node, used-an-ordering)
states, so the reported
cycle is short and every
constraint on it gets a label
Minimal core
QuickXplain: drop any one
listed require and the rest hold
groups appear in it like requires
Proved or budgeted?
proved for every candidate
position this layout considers,
or a probe ran out of checks:
the core may be an artifact
of --budget, and a larger one is suggested
no layout fits makes the same distinction
Help printed
remove one, turn it into
prefer, or use a larger page;
the default 7pt floor can be lowered with font


Diagnostics
rustc style, at the source lines that caused them
fitd explain CODE prints the long form
--short prints one line each
E00xx: front end
E0001 unknown node, with a did-you-mean by edit distance
E0002 syntax error
E0003 node or style declared twice; points at both
E0004 unknown style
E0005 attribute not valid on this statement
E0006 wrong value type or unit
E0007 constraint names a node twice, or align with one node
E0008 contradictory hard orderings, found statically
E0009 unknown layout, relation, side, place, shape, or flag
E0010 two groups share some nodes without one holding the other
E0011 include cycle or unreadable file
E0012 macro error: unknown variable or template, argument count, type, division by zero, nesting too deep
E0013 macro expansion ran past --expand-budget steps
W02xx: warnings
never block output
W0201 link through a node
W0202 missing glyph
W0203 unused style
W0204 repeated setting
W0205 unmet preference
W0206 layered crossing
W0207 links crossing or running together
E01xx: layout
E0101 checker: overlap
E0102 checker: edge through node
E0103 checker: crossing
E0104 checker: off page
E0105 checker: text overflow
E0106 checker: font too small
E0107 checker: child nearer the center
E0108 layout breaks a require or group; a solver bug
E0109 edge label covered, overlapping, or clipped; a solver bug
E0110 minimal set of requires that cannot hold together